

Indiana Repeater Council Membership Portal

Test Plan - Version 2.2

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1. Introduction

1.1 Purpose

This test plan provides comprehensive testing procedures for the IRC Membership Portal to ensure all features work correctly before deployment to production.

1.2 Scope

This plan covers: - User authentication (login, logout, password recovery) - Member CRUD operations (Create, Read, Update, Delete) - Admin-specific features - Status badge logic - Expiration notifications - PDF exports - Email functionality - Data validation - Security controls

1.3 Test Levels

- **Unit Testing:** Individual component validation
- **Integration Testing:** Multi-component workflows
- **System Testing:** End-to-end scenarios
- **Regression Testing:** Verify existing features still work after changes

1.4 Test Environment Requirements

- Docker containers running (clubledger_db, clubledger_web)
- Valid .env configuration
- Test email account accessible

- Sample member data loaded
 - Admin account credentials available
-

2. Test Environment Setup

2.1 Prerequisites Checklist

Docker Compose running (docker compose ps shows healthy containers)
 Database accessible (docker exec clubledger_db mariadb -u root -p)
 Web interface accessible (http://[host]:5000)
 SMTP credentials configured in .env
 Test email account monitored (for receiving emails)
 Browser with dev tools available
 Database backup created before testing

2.2 Test Data Setup

Create test accounts:

```

-- Run in database before testing
USE clubledger_db;

-- Test admin account
INSERT INTO members (call_sign, password_hash, email, name, paid_thru, member_type, is_admin)
VALUES ('TEST1ADM', '[hashed_password]', 'testadmin@test.com', 'Test Admin', '2027', 'FULL', 1);

-- Test regular member (active)
INSERT INTO members (call_sign, password_hash, email, name, paid_thru, member_type, is_admin)
VALUES ('TEST2MEM', '[hashed_password]', 'testmember@test.com', 'Test Member Active', '2027', 'FULL', 0);

-- Test expiring member
INSERT INTO members (call_sign, password_hash, email, name, paid_thru, member_type, is_admin)
VALUES ('TEST3EXP', '[hashed_password]', 'testexpiring@test.com', 'Test Member Expiring', '2026', 'FULL', 0);

-- Test expired member
INSERT INTO members (call_sign, password_hash, email, name, paid_thru, member_type, is_admin)
VALUES ('TEST4OLD', '[hashed_password]', 'testexpired@test.com', 'Test Member Expired', '2025', 'ASSOCIATE', 0);

-- Test member without email
INSERT INTO members (call_sign, password_hash, name, paid_thru, member_type, is_admin)
VALUES ('TEST5NOE', '[hashed_password]', 'Test No Email', '2027', 'LIFE', 0);
  
```

3. Pre-Test Data Preparation

3.1 Verify Database Schema

```

-- Check all required fields exist
DESCRIBE members;

-- Verify these columns exist:
-- id, call_sign, password_hash, email, name, primary_rep, rep_call,
  
```

```
-- address, city, state, zip, telephone, paid_thru, member_type,  
-- is_admin, admin_comments, expiration_status, expiration_notice_sent,  
-- created_at, updated_at
```

3.2 Verify File Structure

```
app/app.py exists  
app/templates/*.html all exist  
.env contains all required variables  
check_expirations.py exists  
backup scripts exist
```

4. Authentication & Security Tests

4.1 Login - Valid Credentials

Test ID: AUTH-001

Priority: Critical

Steps: 1. Navigate to portal login page 2. Enter valid call sign (uppercase): TEST1ADM 3. Enter correct password 4. Click "Login"

Expected Results: - ✓ Redirects to dashboard - ✓ Session cookie created - ✓ User sees member list (if admin) or own profile (if member) - ✓ Logout link visible in header

Status: ☐ Pass ☐ Fail ☐ Blocked

4.2 Login - Invalid Password

Test ID: AUTH-002

Priority: Critical

Steps: 1. Navigate to login page 2. Enter valid call sign: TEST1ADM 3. Enter incorrect password 4. Click "Login"

Expected Results: - ✓ Stays on login page - ✓ Error message displayed: "Invalid call sign or password" - ✓ No session created - ✓ Password field cleared

Status: ☐ Pass ☐ Fail ☐ Blocked

4.3 Login - Non-Existent User

Test ID: AUTH-003

Priority: High

Steps: 1. Navigate to login page 2. Enter non-existent call sign: FAKE123 3. Enter any password 4. Click "Login"

Expected Results: - ✓ Stays on login page - ✓ Error message: "Invalid call sign or password" - ✓ No information leaked about whether user exists

Status: ☐ Pass ☐ Fail ☐ Blocked

4.4 Login - Case Sensitivity

Test ID: AUTH-004

Priority: Medium

Steps: 1. Navigate to login page 2. Enter call sign in lowercase: test1adm 3. Enter correct password 4. Click "Login"

Expected Results: - ✓ Login succeeds (call signs should be case-insensitive) - ✓ System internally converts to uppercase

Status: ☐ Pass ☐ Fail ☐ Blocked

4.5 Password Recovery - Valid Email

Test ID: AUTH-005

Priority: Critical

Steps: 1. Navigate to login page 2. Click "Password Recovery" 3. Enter valid email: testadmin@test.com 4. Click "Send Recovery Link" 5. Check email inbox

Expected Results: - ✓ Success message: "Password reset email sent" - ✓ Email received within 2 minutes - ✓ Email contains reset link - ✓ Link is valid (not expired) - ✓ Link format: /reset-password/[token]

Status: ☐ Pass ☐ Fail ☐ Blocked

4.6 Password Recovery - Invalid Email

Test ID: AUTH-006

Priority: Medium

Steps: 1. Navigate to Password Recovery page 2. Enter email not in system: fake@test.com 3. Click "Send Recovery Link"

Expected Results: - ✓ Success message shown (don't leak info about valid emails) - ✓ No email sent - ✓ No error logged in application

Status: ☐ Pass ☐ Fail ☐ Blocked

4.7 Password Reset - Valid Token

Test ID: AUTH-007

Priority: Critical

Steps: 1. Complete AUTH-005 to get reset link 2. Click link from email 3. Enter new password: TestPass123! 4. Confirm password: TestPass123! 5. Click "Reset Password" 6. Login with new password

Expected Results: - ✓ Redirects to reset page with token - ✓ Password fields visible - ✓ Success message after reset - ✓ Can login with new password - ✓ Old password no longer works

Status: ☐ Pass ☐ Fail ☐ Blocked

4.8 Password Reset - Expired Token

Test ID: AUTH-008

Priority: Medium

Steps: 1. Generate password reset link 2. Wait 25 hours (or manually expire in database) 3. Click expired link

Expected Results: - ✓ Error message: "Reset link expired" - ✓ Option to request new link - ✓ Cannot reset password with old token

Status: ☐ Pass ☐ Fail ☐ Blocked

4.9 Session Management - Logout

Test ID: AUTH-009

Priority: High

Steps: 1. Login as TEST1ADM 2. Navigate to dashboard 3. Click "Logout" link 4. Verify logout 5. Try to access dashboard directly

Expected Results: - ✓ Redirects to login page - ✓ Session cookie deleted - ✓ Cannot access protected pages - ✓ Attempting to access /dashboard redirects to login

Status: ☐ Pass ☐ Fail ☐ Blocked

4.10 Session Management - Timeout

Test ID: AUTH-010

Priority: Medium

Steps: 1. Login as TEST1ADM 2. Wait 24 hours (or manually expire session) 3. Try to perform action

Expected Results: - ✓ Session expires after 24 hours - ✓ Redirects to login - ✓ Must re-authenticate

Status: ☐ Pass ☐ Fail ☐ Blocked

4.11 Authorization - Admin-Only Access

Test ID: AUTH-011

Priority: Critical

Steps: 1. Login as regular member (TEST2MEM) 2. Try to access admin URLs: - /admin/add-member - /profile/[other_user_id] - /admin/delete-member/[id] - /admin/export-pdf

Expected Results: - ✓ Access denied / redirected - ✓ Error message: "Admin access required" - ✓ No data leaked

Status: ☐ Pass ☐ Fail ☐ Blocked

4.12 Authorization - Self-Profile Only

Test ID: AUTH-012

Priority: High

Steps: 1. Login as TEST2MEM (user_id = X) 2. Try to access /profile/[different_user_id]

Expected Results: - ✓ Access denied - ✓ Error: "You do not have permission" - ✓ Can only access own profile (/profile/X)

Status: ☐ Pass ☐ Fail ☐ Blocked

5. Member Management Tests

5.1 Add Member - Complete Valid Data

Test ID: MEMBER-001

Priority: Critical

Steps: 1. Login as admin (TEST1ADM) 2. Click "Add Member" 3. Fill all fields: - Call Sign: NEWADD1 - Email: newadd1@test.com - Name: New Member Test - Address: 123 Test St - City: Indianapolis - State: IN - ZIP: 46220 - Telephone: 317-555-1234 - Member Type: FULL - Paid Through: 2027 - Admin Comments: "Test member - paid \$25 check #999" 4. Check "Grant Admin Privileges" (leave unchecked) 5. Click "Add Member"

Expected Results: - ✓ Success message: "Member NEWADD1 added successfully! Password reset email sent..." - ✓ Redirects to dashboard - ✓ Member visible in member list - ✓ Email sent to newadd1@test.com with password reset link - ✓ Admin comments saved but not visible to member - ✓ Random password generated (not visible to admin)

Status: ☐ Pass ☐ Fail ☐ Blocked

5.2 Add Member - Duplicate Call Sign

Test ID: MEMBER-002

Priority: High

Steps: 1. Login as admin 2. Click "Add Member" 3. Enter call sign that already exists: TEST1ADM 4. Fill other required fields 5. Click "Add Member"

Expected Results: - ✓ Error message: "Call sign or email already exists" - ✓ Member not added - ✓ Form retains entered data - ✓ Database unchanged

Status: ☐ Pass ☐ Fail ☐ Blocked

5.3 Add Member - Duplicate Email

Test ID: MEMBER-003

Priority: High

Steps: 1. Login as admin 2. Click "Add Member" 3. Enter unique call sign but duplicate email: testadmin@test.com 4. Fill other fields 5. Click "Add Member"

Expected Results: - ✓ Error message: "Call sign or email already exists" - ✓ Member not added

Status: ☐ Pass ☐ Fail ☐ Blocked

5.4 Add Member - Missing Required Fields

Test ID: MEMBER-004

Priority: Medium

Steps: 1. Login as admin 2. Click "Add Member" 3. Leave Call Sign empty 4. Fill email and name 5. Try to submit

Expected Results: - ✓ Browser validation prevents submit - ✓ Required field highlighted - ✓ Error message shown

Status: ☐ Pass ☐ Fail ☐ Blocked

5.5 Add Member - Without Email

Test ID: MEMBER-005

Priority: Medium

Steps: 1. Login as admin 2. Click "Add Member" 3. Fill call sign and name 4. Leave email blank 5. Submit

Expected Results: - ✓ Warning message: "Member added, but no email provided" - ✓ Member created in database - ✓ No password reset email sent - ✓ Admin must manually assist member with password

Status: ☐ Pass ☐ Fail ☐ Blocked

5.6 Add Member - Admin Comments Character Limit

Test ID: MEMBER-006

Priority: Low

Steps: 1. Login as admin 2. Click "Add Member" 3. Fill required fields 4. In Admin Comments, paste 600 characters of text 5. Submit

Expected Results: - ✓ Character counter shows 500/500 - ✓ Only first 500 characters saved - ✓ Form prevents entry beyond 500 chars OR truncates

Status: ☐ Pass ☐ Fail ☐ Blocked

5.7 View Member List - Admin View

Test ID: MEMBER-007

Priority: High

Steps: 1. Login as admin (TEST1ADM) 2. View dashboard

Expected Results: - ✓ All members visible - ✓ Sortable columns (click header to sort) - ✓ Status badges show correct colors: - Green for paid 2027+ - Yellow for paid 2026 - Red for paid 2025- - ✓ Action buttons visible for each member: - Send Update Notice (envelope) - Reset Password (key) - Delete (trash) - ✓ Call signs are blue clickable links

Status: ☐ Pass ☐ Fail ☐ Blocked

5.8 View Member List - Regular Member View

Test ID: MEMBER-008

Priority: High

Steps: 1. Login as regular member (TEST2MEM) 2. View dashboard

Expected Results: - ✓ Only sees own profile - ✓ No action buttons visible - ✓ Cannot see other members - ✓ "My Profile" heading instead of "Member Management"

Status: ☐ Pass ☐ Fail ☐ Blocked

5.9 Delete Member - Confirmation Required

Test ID: MEMBER-009

Priority: Critical

Steps: 1. Login as admin 2. Navigate to dashboard 3. Click delete (trash icon) for TEST4OLD 4. Observe confirmation dialog 5. Click "Cancel" 6. Verify member still exists

Expected Results: - ✓ Confirmation dialog appears - ✓ Dialog warns: "Are you sure? This cannot be undone" - ✓ Cancel button prevents deletion - ✓ Member still in database

Status: ☐ Pass ☐ Fail ☐ Blocked

5.10 Delete Member - Successful Deletion

Test ID: MEMBER-010

Priority: High

Steps: 1. Login as admin 2. Navigate to dashboard 3. Click delete for TEST4OLD 4. Confirm deletion 5. Verify removal

Expected Results: - ✓ Confirmation dialog appears - ✓ Click "Confirm" - ✓ Success message: "Member deleted" - ✓ Member removed from list - ✓ Database record deleted - ✓ Cannot login with deleted account

Status: ☐ Pass ☐ Fail ☐ Blocked

5.11 Delete Member - Cannot Delete Self

Test ID: MEMBER-011

Priority: High

Steps: 1. Login as admin (TEST1ADM) 2. Navigate to dashboard 3. Attempt to delete own account

Expected Results: - ✓ Delete button not visible for own account - ✓ If manually accessed via URL: Error message - ✓ "You cannot delete your own account"

Status: ☐ Pass ☐ Fail ☐ Blocked

6. Profile Management Tests

6.1 Edit Own Profile - Regular Member

Test ID: PROFILE-001

Priority: High

Steps: 1. Login as TEST2MEM 2. View profile 3. Update editable fields: - Email: newemail@test.com - Address: 456 New St - City: Carmel - Telephone: 317-555-9999 4. Click "Save Changes"

Expected Results: - ✓ Success message: "Profile updated" - ✓ Changes saved in database - ✓ Cannot edit: - Call Sign (disabled/grayed) - Member Type (disabled) - Paid Through (disabled) - Admin Comments (not visible)

Status: ☐ Pass ☐ Fail ☐ Blocked

6.2 Edit Own Profile - Admin Cannot Demote Self

Test ID: PROFILE-002

Priority: Critical

Steps: 1. Login as admin (TEST1ADM) 2. Edit own profile 3. Uncheck "Grant Admin Privileges" 4. Save

Expected Results: - ✓ Admin checkbox should not be changeable for own profile - ✓ OR if changeable, saving preserves admin status - ✓ Cannot accidentally demote self

Status: ☐ Pass ☐ Fail ☐ Blocked

6.3 Edit Other Member - Admin Full Access

Test ID: PROFILE-003

Priority: High

Steps: 1. Login as admin (TEST1ADM) 2. Click TEST2MEM call sign 3. Update all fields including: - Member Type: ASSOCIATE - Paid Through: 2028 - Admin Comments: "Updated by admin" 4. Check "Grant Admin Privileges" 5. Save

Expected Results: - ✓ All fields editable by admin - ✓ Changes saved - ✓ Member Type updated - ✓ Paid Through updated - ✓ Admin status granted - ✓ Admin Comments saved

Status: ☐ Pass ☐ Fail ☐ Blocked

6.4 Change Call Sign - Admin Only

Test ID: PROFILE-004

Priority: High

Steps: 1. Login as admin 2. Edit TEST3EXP 3. Change call sign from TEST3EXP to TEST3NEW 4. Save

Expected Results: - ✓ Call sign updated in database - ✓ Member can login with TEST3NEW - ✓ Cannot login with TEST3EXP anymore - ✓ If member logged in, they're logged out - ✓ Flash message: "Call sign updated, member must re-login"

Status: ☐ Pass ☐ Fail ☐ Blocked

6.5 Change Call Sign - Duplicate Prevention

Test ID: PROFILE-005

Priority: High

Steps: 1. Login as admin 2. Edit TEST2MEM 3. Try to change call sign to TEST1ADM (existing) 4. Save

Expected Results: - ✓ Error message: "Call sign TEST1ADM is already in use" - ✓ Change not saved - ✓ Original call sign preserved

Status: ☐ Pass ☐ Fail ☐ Blocked

6.6 Change Call Sign - Self (Admin)

Test ID: PROFILE-006

Priority: Medium

Steps: 1. Login as TEST1ADM 2. Edit own profile 3. Change call sign from TEST1ADM to TEST1NEW 4. Save

Expected Results: - ✓ Call sign updated - ✓ Immediately logged out - ✓ Flash: "Call sign updated to TEST1NEW. Please login with new call sign" - ✓ Must re-login with TEST1NEW

Status: ☐ Pass ☐ Fail ☐ Blocked

6.7 Change Password - Valid Current Password

Test ID: PROFILE-007

Priority: High

Steps: 1. Login as TEST2MEM 2. Scroll to Security section 3. Click "Change Password" 4. Enter current password 5. Enter new password: NewPass123! 6. Confirm new password: NewPass123! 7. Click "Update Password"

Expected Results: - ✓ Success message: "Password updated" - ✓ Can logout and re-login with new password - ✓ Old password no longer works

Status: ☐ Pass ☐ Fail ☐ Blocked

6.8 Change Password - Wrong Current Password

Test ID: PROFILE-008

Priority: Medium

Steps: 1. Login as TEST2MEM 2. Try to change password 3. Enter incorrect current password 4. Enter new password 5. Submit

Expected Results: - ✓ Error: "Current password incorrect" - ✓ Password not changed - ✓ Can still login with original password

Status: ☐ Pass ☐ Fail ☐ Blocked

6.9 Change Password - Mismatch Confirmation

Test ID: PROFILE-009

Priority: Low

Steps: 1. Login as TEST2MEM 2. Change password form 3. Enter correct current password 4. New password: Test123! 5. Confirm password: Test456! (different) 6. Submit

Expected Results: - ✓ Error: "Passwords do not match" - ✓ Password not changed

Status: ☐ Pass ☐ Fail ☐ Blocked

6.10 Admin Comments - Visibility Test

Test ID: PROFILE-010

Priority: High

Steps: 1. Login as admin 2. Edit TEST2MEM profile 3. Add admin comment: "Paid \$25 on 02/05/26" 4. Save 5. Logout 6. Login as TEST2MEM 7. View own profile

Expected Results: - ✓ Admin can see and edit comments field - ✓ Regular member CANNOT see admin comments field - ✓ Comments saved in database - ✓ Field label clearly says "Admin Only"

Status: ☐ Pass ☐ Fail ☐ Blocked

7. Admin Features Tests

7.1 Send Update Notification - Valid Member

Test ID: ADMIN-001

Priority: High

Steps: 1. Login as admin 2. Dashboard view 3. Click envelope icon for TEST2MEM 4. Confirm sending 5. Check member's email

Expected Results: - ✓ Confirmation dialog appears - ✓ Success message after sending - ✓ Email received within 2 minutes - ✓ Subject: "Your Indiana Repeater Council membership record has been updated" - ✓ Contains link to portal

Status: ☐ Pass ☐ Fail ☐ Blocked

7.2 Send Update Notification - No Email

Test ID: ADMIN-002

Priority: Medium

Steps: 1. Login as admin 2. Try to send update notice to TEST5NOE (no email)

Expected Results: - ✓ Button not visible OR - ✓ Error message: "Member has no email address" - ✓ No email sent

Status: ☐ Pass ☐ Fail ☐ Blocked

7.3 Send Update Notification - Cannot Send to Self

Test ID: ADMIN-003

Priority: Low

Steps: 1. Login as admin (TEST1ADM) 2. View dashboard 3. Find own entry

Expected Results: - ✓ Envelope button not visible for own account - ✓ Cannot send update notice to self

Status: ☐ Pass ☐ Fail ☐ Blocked

7.4 Reset Password (Admin Initiated) - Valid Email

Test ID: ADMIN-004

Priority: High

Steps: 1. Login as admin 2. Click key icon for TEST2MEM 3. Confirm email address 4. Send reset 5. Check member's email

Expected Results: - ✓ Confirmation dialog shows email address - ✓ Success message - ✓ Password reset email sent to member - ✓ Member can use link to reset password

Status: ☐ Pass ☐ Fail ☐ Blocked

7.5 Reset Password (Admin Initiated) - No Email

Test ID: ADMIN-005

Priority: Medium

Steps: 1. Login as admin 2. Try to reset password for TEST5NOE (no email)

Expected Results: - ✓ Button not visible OR - ✓ Error: "Member has no email address"

Status: ☐ Pass ☐ Fail ☐ Blocked

7.6 Sortable Columns - Click to Sort

Test ID: ADMIN-006

Priority: Medium

Steps: 1. Login as admin 2. View dashboard with multiple members 3. Click "Call Sign" column header 4. Click "Name" column header 5. Click "Paid Through" column header

Expected Results: - ✓ Clicking header sorts ascending - ✓ Clicking again sorts descending - ✓ Visual indicator (arrow) shows sort direction - ✓ All columns sortable

Status: ☐ Pass ☐ Fail ☐ Blocked

7.7 Clickable Call Signs

Test ID: ADMIN-007

Priority: Medium

Steps: 1. Login as admin 2. View dashboard 3. Click on any call sign (blue link)

Expected Results: - ✓ Call signs displayed as blue clickable links - ✓ Clicking opens profile edit page - ✓ Correct member profile loads

Status: ☐ Pass ☐ Fail ☐ Blocked

8. Status & Notification Tests

8.1 Status Badge - Active (Green)

Test ID: STATUS-001

Priority: High

Steps: 1. Login as admin 2. View member with paid_thru = 2027 (future year)

Expected Results: - ✓ Status badge shows "Active" - ✓ Badge color is GREEN - ✓ Tooltip or text: "Paid through 2027"

Status: ☐ Pass ☐ Fail ☐ Blocked

8.2 Status Badge - Expiring (Yellow)

Test ID: STATUS-002

Priority: High

Steps: 1. Login as admin 2. View member with paid_thru = 2026 (current year)

Expected Results: - ✓ Status badge shows "Expiring" - ✓ Badge color is YELLOW - ✓ Shows all year (not just Nov/Dec)

Status: ☐ Pass ☐ Fail ☐ Blocked

8.3 Status Badge - Expired (Red)

Test ID: STATUS-003

Priority: High

Steps: 1. Login as admin 2. View member with paid_thru = 2025 (past year)

Expected Results: - ✓ Status badge shows "Expired" - ✓ Badge color is RED

Status: ☐ Pass ☐ Fail ☐ Blocked

8.4 Status Calculation - Boundary Test

Test ID: STATUS-004

Priority: Medium

Steps: 1. Create test member with paid_thru = current year 2. Verify status is "Expiring" (yellow) 3. Update paid_thru = current year + 1 4. Verify status changes to "Active" (green) 5. Update paid_thru = current year - 1 6. Verify status changes to "Expired" (red)

Expected Results: - ✓ Status updates immediately after paid_thru change - ✓ Correct color coding - ✓ Database expiration_status field updates

Status: ☐ Pass ☐ Fail ☐ Blocked

8.5 Expiration Notification - Manual Test Run

Test ID: NOTIFY-001

Priority: High

Steps: 1. SSH to server 2. Run: ./run_expiration_check.sh 3. Monitor console output 4. Check admin email

Expected Results: - ✓ Script runs without errors - ✓ Console shows member count - ✓ Shows status changes detected - ✓ Admin summary email received - ✓ Email contains: - Total notifications sent - Status changes breakdown - Failed notifications (if any) - Members without email

Status: ☐ Pass ☐ Fail ☐ Blocked

8.6 Expiration Notification - Status Change Detection

Test ID: NOTIFY-002

Priority: High

Steps: 1. Check TEST3EXP current status in database 2. Update paid_thru to trigger status change 3. Run notification script 4. Check member's email

Expected Results: - ✓ Status change detected - ✓ Notification email sent to member - ✓ expiration_status updated in database - ✓ expiration_notice_sent date set to today - ✓ Admin receives summary

Status: ☐ Pass ☐ Fail ☐ Blocked

8.7 Expiration Notification - No Duplicate Sends

Test ID: NOTIFY-003

Priority: High

Steps: 1. Run notification script first time 2. Note which members notified 3. Run script again immediately (no changes) 4. Check email

Expected Results: - ✓ First run: Notifications sent - ✓ Second run: No notifications sent - ✓ Message: "No status changes detected" - ✓ Members don't receive duplicate emails

Status: ☐ Pass ☐ Fail ☐ Blocked

8.8 Expiration Notification - Member Without Email

Test ID: NOTIFY-004

Priority: Medium

Steps: 1. Ensure TEST5NOE has no email 2. Change paid_thru to trigger status change 3. Run notification script 4. Check admin summary

Expected Results: - ✓ Status change detected - ✓ NO email sent (no email address) - ✓ Logged in admin summary as "failed" - ✓ Reason: "NO EMAIL"

Status: ☐ Pass ☐ Fail ☐ Blocked

8.9 Expiration Notification - Email Content

Test ID: NOTIFY-005

Priority: Medium

Steps: 1. Trigger expiration notification 2. Check member email content

Expected Results: - ✓ Subject line appropriate for status - ✓ Body contains: - Member's call sign - Current paid through year - Clear action needed - Link to portal - Contact information - ✓ Professional formatting - ✓ IRC branding/signature

Status: ☐ Pass ☐ Fail ☐ Blocked

9. Export & Reporting Tests

9.1 PDF Export - Generate Report

Test ID: EXPORT-001

Priority: High

Steps: 1. Login as admin 2. Click "Export PDF" button 3. Wait for download 4. Open PDF

Expected Results: - ✓ PDF downloads automatically - ✓ Filename: {ORG_NAME}_Membership_YYYYMMDD_H
- ✓ Opens without errors - ✓ Contains all members

Status: ☐ Pass ☐ Fail ☐ Blocked

9.2 PDF Export - Content Validation

Test ID: EXPORT-002

Priority: High

Steps: 1. Generate PDF export 2. Review contents

Expected Results: - ✓ Header: "Indiana Repeater Council Membership Database" - ✓ Generation date and time - ✓ Total member count - ✓ Table with columns: - Call Sign, Name, Email, City, State - Member Type, Paid Through, Admin - ✓ All rows populated - ✓ Professional formatting - ✓ Landscape orientation - ✓ Alternating row colors - ✓ Footer: Confidentiality notice

Status: ☐ Pass ☐ Fail ☐ Blocked

9.3 PDF Export - Sort Order

Test ID: EXPORT-003

Priority: Medium

Steps: 1. Generate PDF export 2. Check member order

Expected Results: - ✓ Members sorted by LAST NAME (not call sign) - ✓ Alphabetical A-Z - ✓
Example: "John Smith" before "Mary Wilson"

Status: ☐ Pass ☐ Fail ☐ Blocked

9.4 PDF Export - Large Dataset

Test ID: EXPORT-004

Priority: Low

Steps: 1. Add 100+ test members 2. Generate PDF export 3. Review

Expected Results: - ✓ All members included - ✓ Multiple pages if needed - ✓ Page numbers visible - ✓ Headers repeat on each page - ✓ No truncation

Status: ☐ Pass ☐ Fail ☐ Blocked

9.5 PDF Export - Non-Admin Cannot Access

Test ID: EXPORT-005

Priority: Medium

Steps: 1. Login as regular member 2. Try to access /admin/export-pdf directly

Expected Results: - ✓ Access denied - ✓ Redirected or error message - ✓ PDF not generated

Status: ☐ Pass ☐ Fail ☐ Blocked

9.6 Database Backup - Automated Email

Test ID: EXPORT-006

Priority: Medium

Steps: 1. Run backup script: ./backup_and_email.sh 2. Check admin emails

Expected Results: - ✓ Backup creates ZIP file - ✓ ZIP contains: - SQL dump - CSV export - ✓
Email sent to all admin addresses - ✓ Email has ZIP attached - ✓ Email body contains: - Backup date/time - Member count - Restore instructions

Status: ☐ Pass ☐ Fail ☐ Blocked

10. Edge Cases & Error Handling

10.1 SQL Injection - Login Form

Test ID: EDGE-001

Priority: Critical

Steps: 1. Try to login with: - Call sign: ' OR '1'='1 - Password: anything 2. Try other SQL injection patterns

Expected Results: - ✓ Login fails - ✓ No SQL error messages - ✓ No unauthorized access - ✓ Queries properly parameterized

Status: ☐ Pass ☐ Fail ☐ Blocked

10.2 XSS - Name Field

Test ID: EDGE-002

Priority: High

Steps: 1. Login as admin 2. Add member with name: <script>alert('XSS')</script> 3. View dashboard

Expected Results: - ✓ Script not executed - ✓ Name displayed as plain text - ✓ HTML escaped properly - ✓ No alert popup

Status: ☐ Pass ☐ Fail ☐ Blocked

10.3 XSS - Admin Comments

Test ID: EDGE-003

Priority: High

Steps: 1. Add member 2. In admin comments: <script>alert('test')</script> 3. View profile

Expected Results: - ✓ Script not executed - ✓ Comments displayed as plain text - ✓ Proper escaping

Status: ☐ Pass ☐ Fail ☐ Blocked

10.4 Long Input Strings - Call Sign

Test ID: EDGE-004

Priority: Low

Steps: 1. Try to add member with 100-character call sign 2. Try to add member with special characters: TEST@#\$\$%

Expected Results: - ✓ Reasonable length limit enforced - ✓ Invalid characters rejected or sanitized - ✓ Helpful error message

Status: ☐ Pass ☐ Fail ☐ Blocked

10.5 Database Connection Loss

Test ID: EDGE-005

Priority: Medium

Steps: 1. Stop database container: `docker stop clubledger_db` 2. Try to access portal 3. Restart database

Expected Results: - ✓ Graceful error message - ✓ No stack trace exposed - ✓ "Database unavailable" message - ✓ Recovers when DB restarted

Status: ☐ Pass ☐ Fail ☐ Blocked

10.6 SMTP Failure - Add Member

Test ID: EDGE-006

Priority: Medium

Steps: 1. Temporarily misconfigure SMTP in `.env` 2. Try to add new member with email 3. Check result

Expected Results: - ✓ Member added to database - ✓ Warning message: "Member added but failed to send email" - ✓ Admin can manually send reset later

Status: ☐ Pass ☐ Fail ☐ Blocked

10.7 Concurrent Edits

Test ID: EDGE-007

Priority: Low

Steps: 1. Login as admin in two browsers 2. Edit same member in both 3. Save from browser 1 4. Save from browser 2

Expected Results: - ✓ Last write wins (acceptable) - ✓ No database corruption - ✓ Both saves complete

Status: ☐ Pass ☐ Fail ☐ Blocked

10.8 Browser Back Button

Test ID: EDGE-008

Priority: Low

Steps: 1. Login 2. Navigate through pages 3. Click browser back button multiple times 4. Try to perform actions

Expected Results: - ✓ Session still valid - ✓ Pages reload properly - ✓ No stale data shown - ✓ CSRF tokens still valid

Status: ☐ Pass ☐ Fail ☐ Blocked

10.9 Special Characters - All Fields

Test ID: EDGE-009

Priority: Medium

Steps: 1. Create member with special chars in all fields: - Name: O'Brien-Smith Jr. - Address: 123½ Main St. Apt #5 - City: Saint-François - Email: test+tag@domain.co.uk

Expected Results: - ✓ All special characters accepted - ✓ Saved correctly - ✓ Display correctly - ✓ No encoding issues

Status: ☐ Pass ☐ Fail ☐ Blocked

10.10 Empty/Null Values

Test ID: EDGE-010

Priority: Medium

Steps: 1. Edit member 2. Clear optional fields (address, phone, etc.) 3. Save

Expected Results: - ✓ Optional fields can be empty - ✓ No errors on NULL values - ✓ Display handles empty fields gracefully

Status: ☐ Pass ☐ Fail ☐ Blocked

11. Performance Tests

11.1 Dashboard Load Time - 100 Members

Test ID: PERF-001

Priority: Medium

Steps: 1. Ensure database has ~100 members 2. Clear browser cache 3. Login as admin 4. Time dashboard load

Expected Results: - ✓ Page loads in < 2 seconds - ✓ No lag scrolling - ✓ All data visible

Status: ☐ Pass ☐ Fail ☐ Blocked

11.2 PDF Generation Time

Test ID: PERF-002

Priority: Low

Steps: 1. Login as admin 2. Time PDF export for 100+ members

Expected Results: - ✓ PDF generates in < 5 seconds - ✓ Download starts immediately

Status: ☐ Pass ☐ Fail ☐ Blocked

11.3 Database Query Performance

Test ID: PERF-003

Priority: Low

Steps: 1. SSH to server 2. Run: ./check_expirations.py 3. Monitor execution time

Expected Results: - ✓ Script completes in < 30 seconds for 100 members - ✓ No timeout errors

Status: ☐ Pass ☐ Fail ☐ Blocked

12. Regression Test Checklist

After any code changes, run this quick checklist:

Login works (admin and regular)
Add member works (with auto-password)
Edit member works
Delete member works (with confirmation)
Status badges show correct colors
PDF export works and sorts by last name
Admin comments visible to admins only
Password reset emails send
Update notifications send
Expiration script runs without errors
All links in navigation work
Logout works properly
Non-admins cannot access admin features

13. Sign-Off

Test Summary

Total Test Cases: ____

Passed: ____

Failed: ____

Blocked: ____

Not Tested: ____

Critical Issues Found

Issue ID	Description	Severity	Status
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Recommendation

- ☐ **PASS** - System ready for production deployment
- ☐ **CONDITIONAL PASS** - Deploy with known issues documented
- ☐ **FAIL** - Critical issues must be resolved before deployment

Sign-Off

Tester Name: _____

Date: _____

Signature: _____

Approver Name: _____

Date: _____

Signature: _____

Appendix A: Test Data Cleanup

After testing, clean up test accounts:

```
DELETE FROM members WHERE call_sign LIKE 'TEST%';  
DELETE FROM members WHERE call_sign = 'NEWADD1';  
DELETE FROM members WHERE call_sign = 'TEST3NEW';  
DELETE FROM members WHERE call_sign = 'TEST1NEW';
```

Appendix B: Known Limitations

- Session timeout is 24 hours (cannot be changed without code modification)
- PDF export limited to ~1000 members (performance degrades beyond that)
- Email delivery depends on SMTP provider reliability
- Backup retention is 30 days (configurable in script)

Appendix C: Browser Compatibility

Test in these browsers: - ☐ Chrome (latest) - ☐ Firefox (latest) - ☐ Safari (latest) - ☐ Edge (latest) - ☐ Mobile Safari (iOS) - ☐ Mobile Chrome (Android)

End of Test Plan